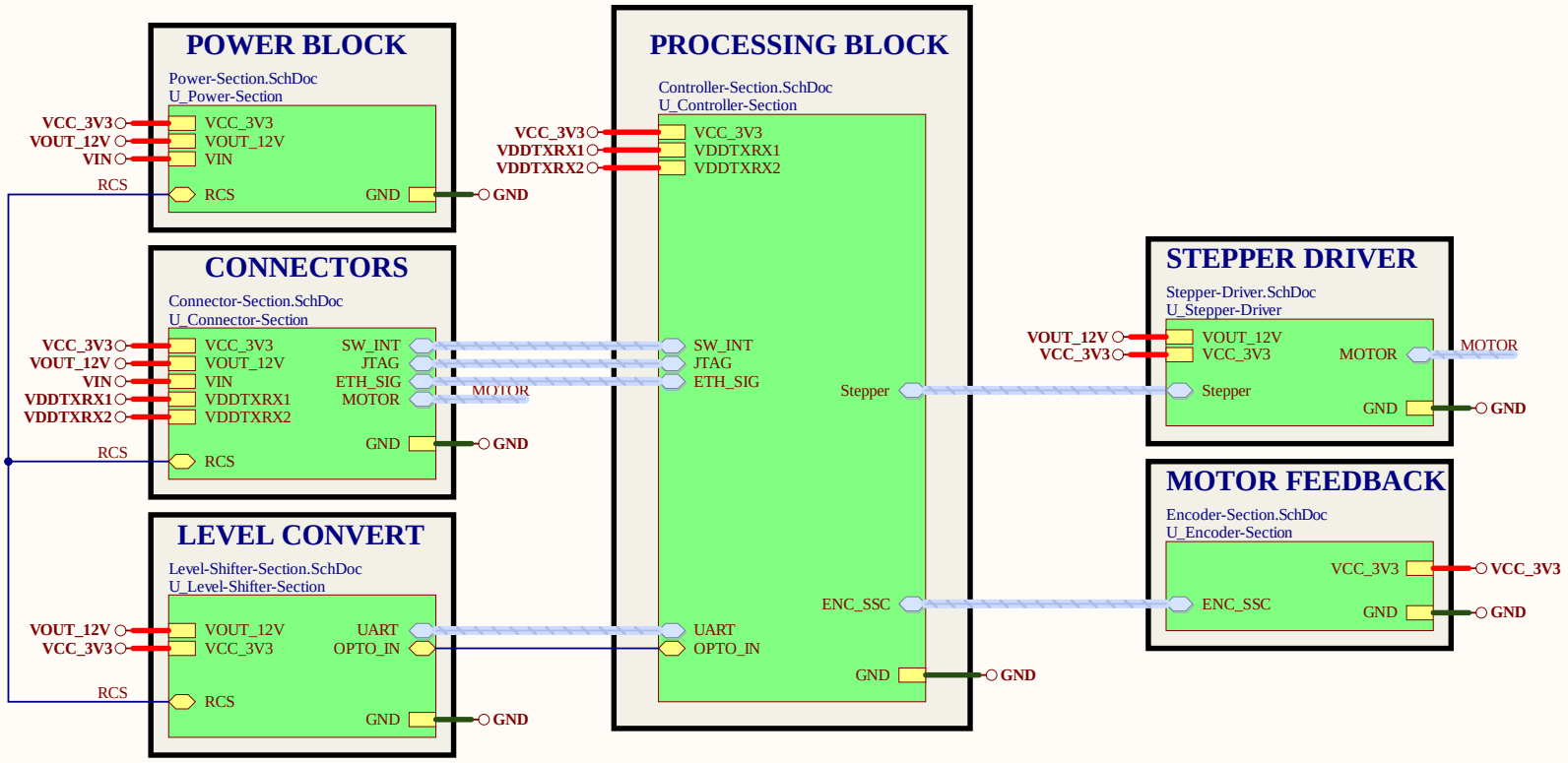
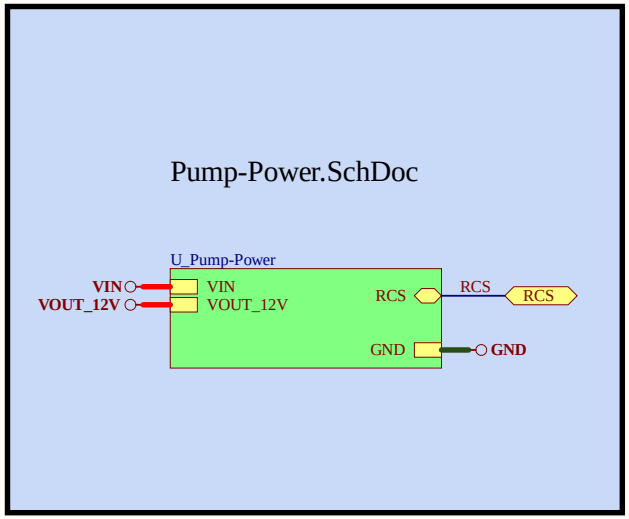
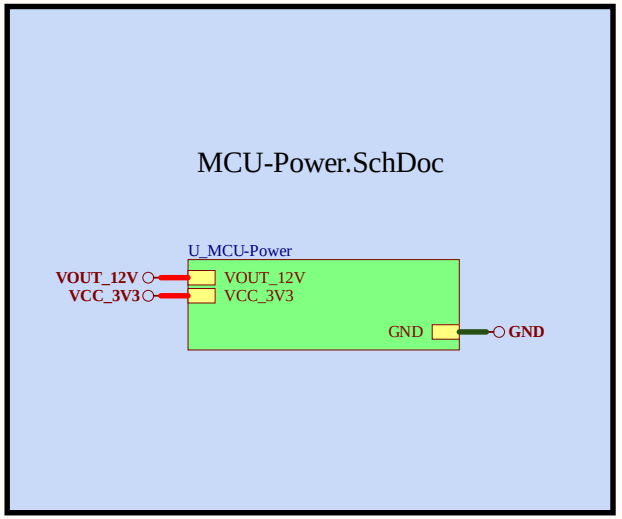
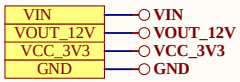
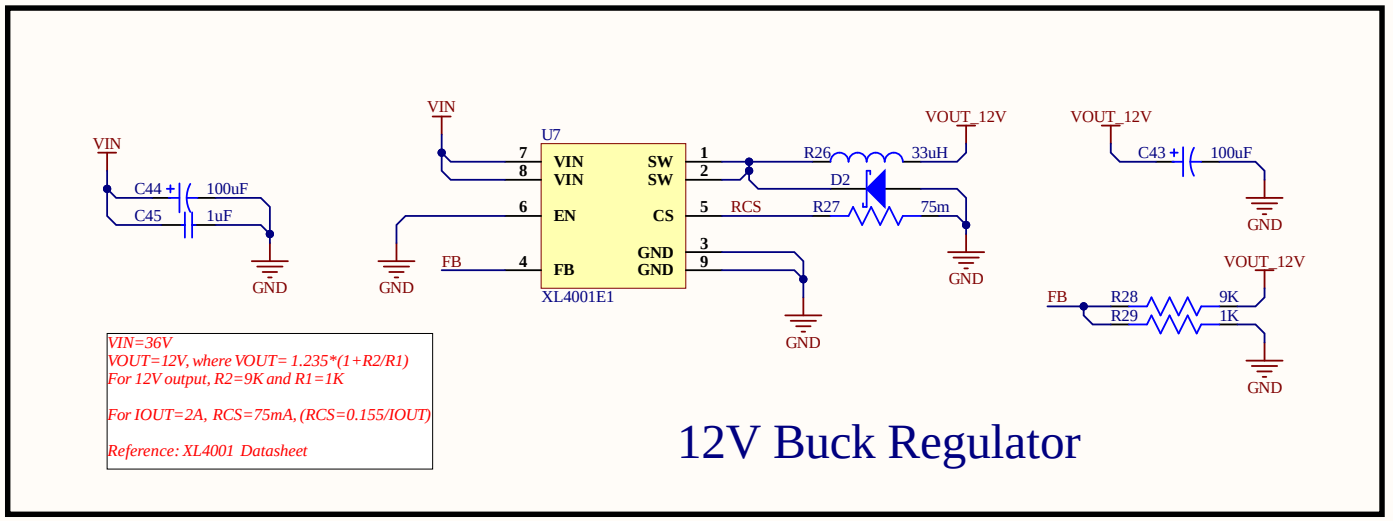

POWER BLOCK
Power-Section.SchDoc
U_Power-Section
VCC_3V3
VOUT_12V
VIN
RCS
GND
GND
CONNECTORS
Connector-Section.SchDoc
U_Connector-Section
SW_INT
JTAG
ETH_SIG
MOTOR
GND
GND
LEVEL CONVERT
Level-Shifter-Section.SchDoc
U_Level-Shifter-Section
UART
OPTO_IN
GND
GND
PROCESSING BLOCK
Controller-Section.SchDoc
U_Controller-Section
VDDTXRX1
VDDTXRX2
SW_INT
JTAG
ETH_SIG
Stepper
ENC_SSC
UART
OPTO_IN
GND
GND
STEPPER DRIVER
Stepper-Driver.SchDoc
U_Stepper-Driver
MOTOR
Stepper
GND
GND
MOTOR FEEDBACK
Encoder-Section.SchDoc
U_Encoder-Section
VCC_3V3
ENC_SSC
GND
VCC_3V3
GND


Pump-Power.SchDoc
U_Pump-Power
VIN
VOUT_12V
VIN
VOUT_12V
RCS
RCS
RCS
GND
GND
GND

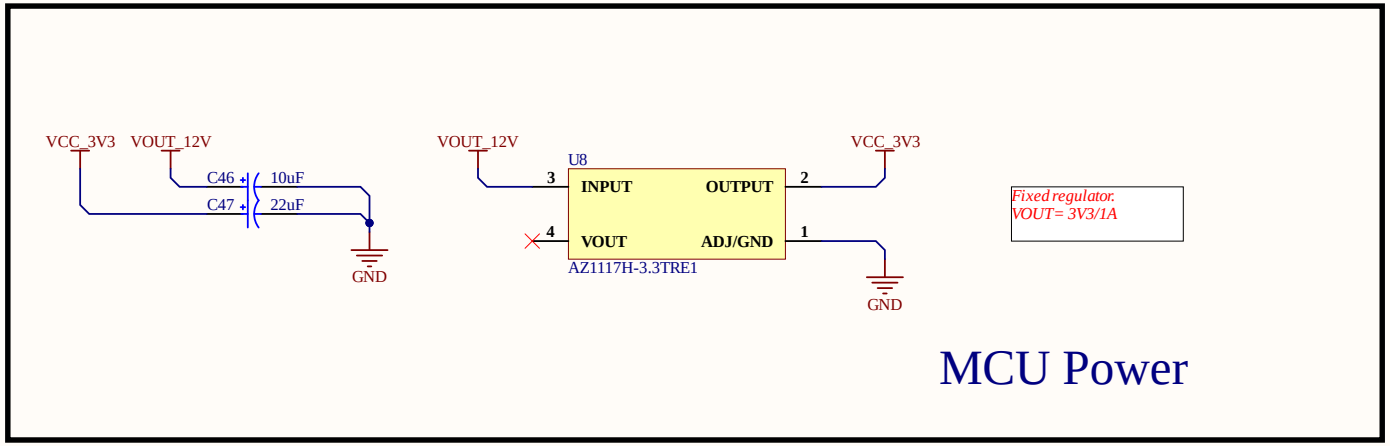

MCU-Power.SchDoc
U_MCU-Power
VOUT_12V
VCC_3V3
VOUT_12V
VCC_3V3
GND
GND


VIN
VOUT_12V
VCC_3V3
GND
VIN
VOUT_12V
VCC_3V3
GND

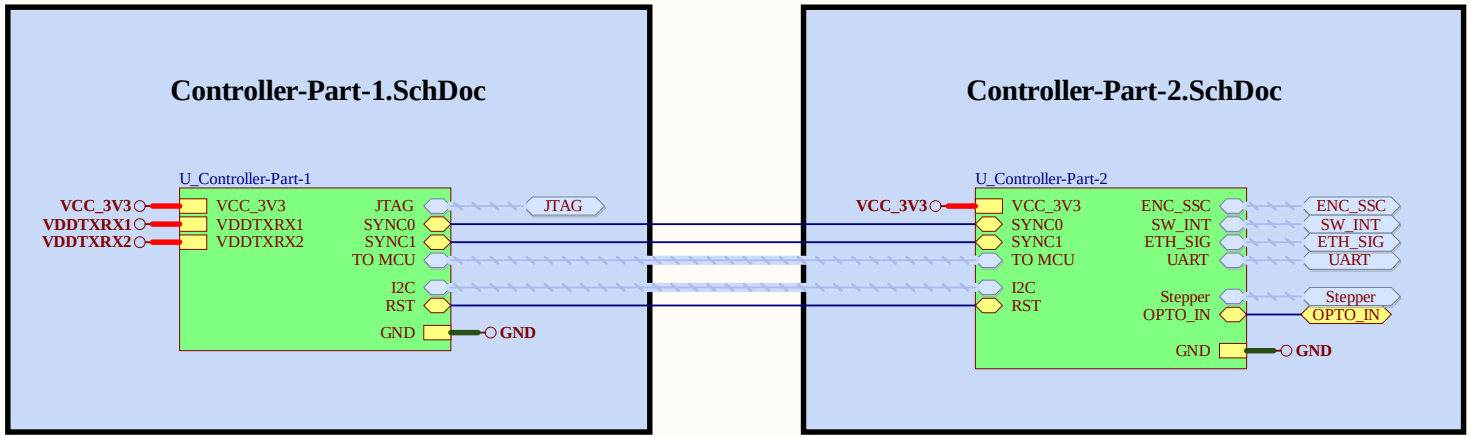
RCS RCS

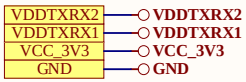


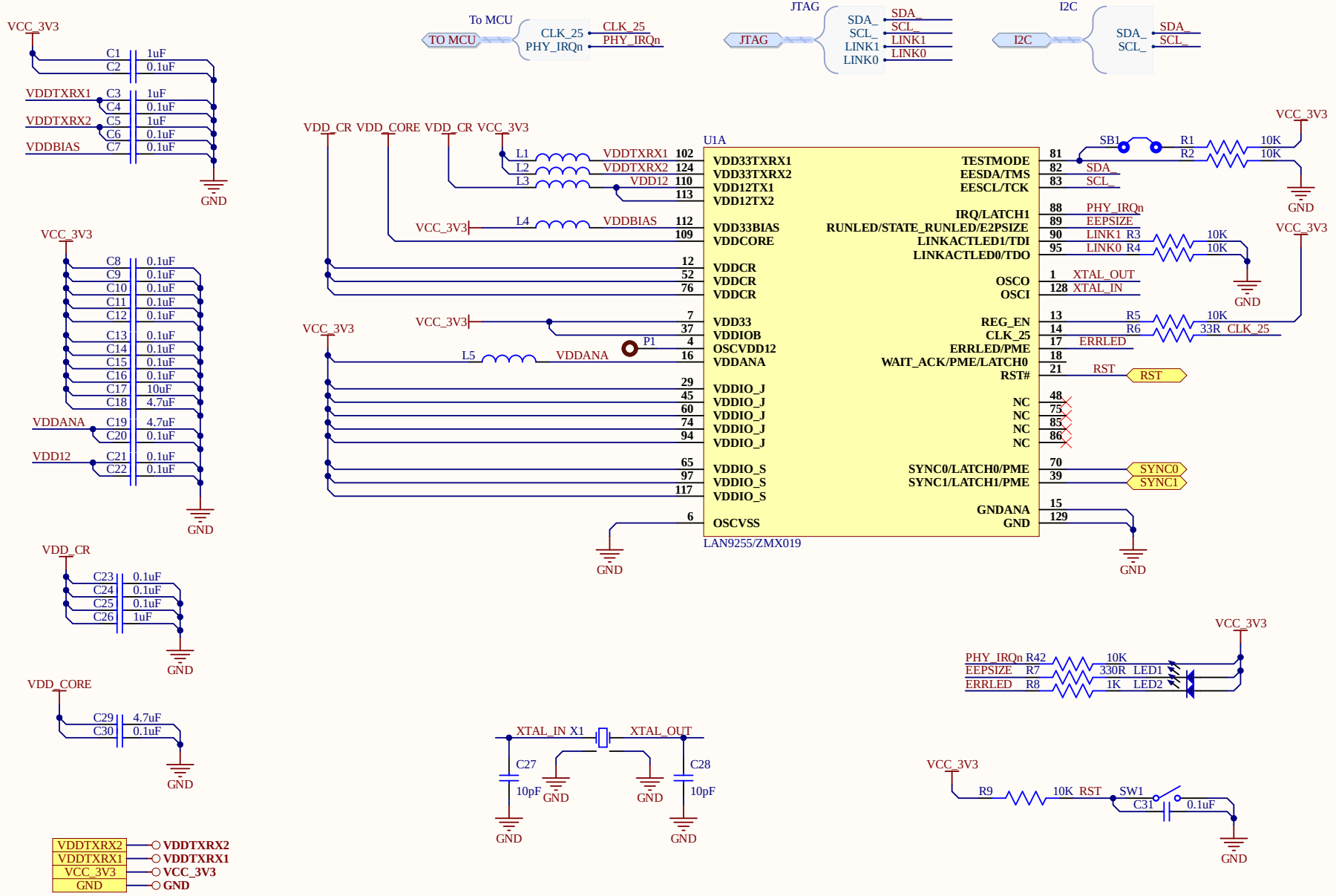
VIN VIN
VOUT_12V VOUT_12V
GND GND

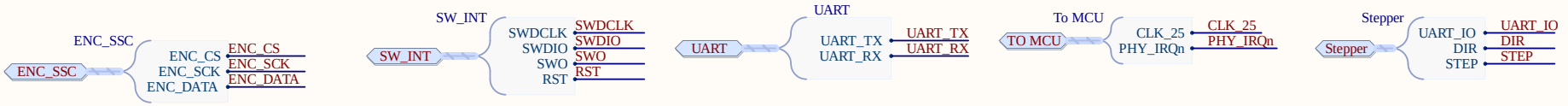


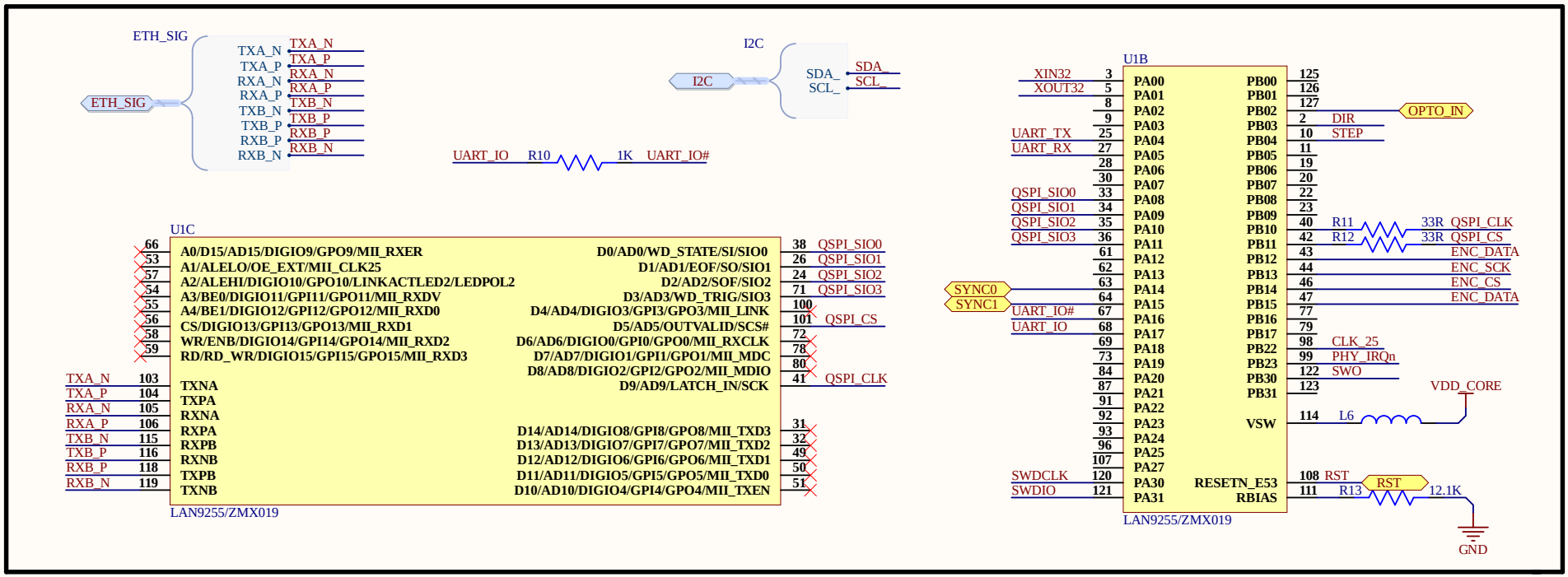
VOUT_12V → VOUT_12V
VCC_3V3 → VCC_3V3
GND → GND

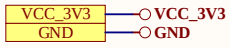

Controller-Part-1.SchDoc
U_Controller-Part-1
VCC_3V3
VDDTXRX1
VDDTXRX2
JTAG
SYNC0
SYNC1
TO MCU
I2C
RST
GND
JTAG
GND
Controller-Part-2.SchDoc
U_Controller-Part-2
VCC_3V3
SYNC0
SYNC1
TO MCU
I2C
RST
ENC_SSC
SW_INT
ETH_SIG
UART
Stepper
OPTO_IN
GND
ENC_SSC
SW_INT
ETH_SIG
UART
Stepper
OPTO_IN
GND

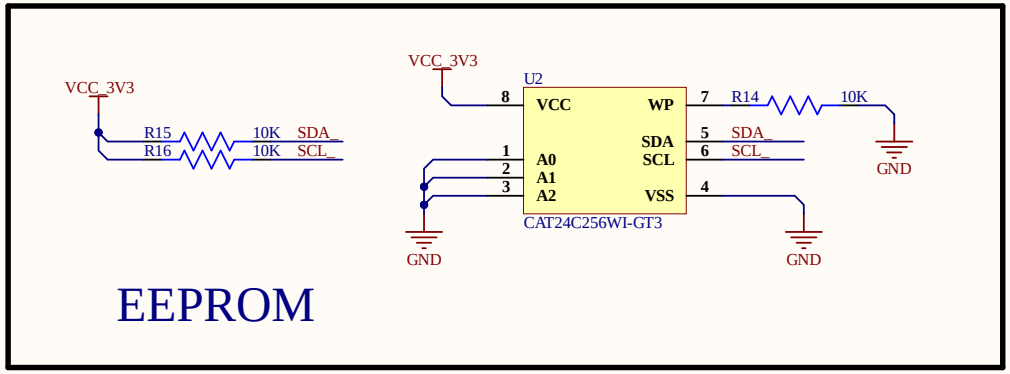

VDDTXRX2
VDDTXRX1
VCC_3V3
GND
VDDTXRX2
VDDTXRX1
VCC_3V3
GND

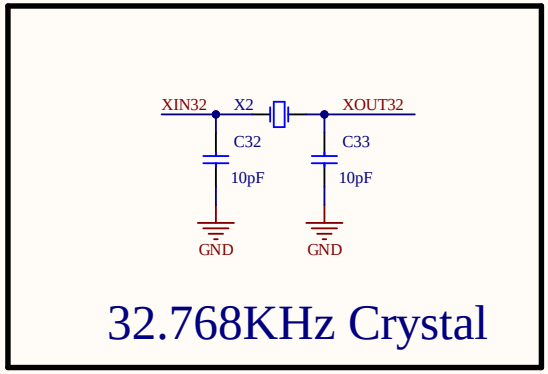

VCC_3V3
C1 1uF
C2 0.1uF
VDDTXRX1 C3 1uF
C4 0.1uF
VDDTXRX2 C5 1uF
C6 0.1uF
VDDBIAS C7 0.1uF
GND
C8 0.1uF
C9 0.1uF
C10 0.1uF
C11 0.1uF
C12 0.1uF
C13 0.1uF
C14 0.1uF
C15 0.1uF
C16 0.1uF
C17 10uF
C18 4.7uF
VDDANA C19 4.7uF
C20 0.1uF
VDD12 C21 0.1uF
C22 0.1uF
VDD_CR
C23 0.1uF
C24 0.1uF
C25 0.1uF
C26 1uF
VDD_CORE
C29 4.7uF
C30 0.1uF
VDDTXRX2
VDDTXRX1
VCC_3V3
GND
To MCU
TO MCU
CLK_25
PHY_IRQn
JTAG
SDA_
SCL_
LINK1
LINK0
I2C
VDD_CR VDD_CORE VDD_CR VCC_3V3
L1 VDDTXRX1 102
L2 VDDTXRX2 124
L3 VDD12 110
113
VCC_3V3 L4 VDDBIAS 112
109
12
52
76
VCC_3V3
7
37
P1 4
L5 VDDANA 16
29
45
60
74
94
65
97
117
6
U1A
VDD33TXRX1
VDD33TXRX2
VDD12TX1
VDD12TX2
VDD33BIAS
VDDCORE
VDDCR
VDDCR
VDDCR
VDD33
VDDIOB
OSCVDD12
VDDANA
VDDIO_J
VDDIO_J
VDDIO_J
VDDIO_J
VDDIO_J
VDDIO_S
VDDIO_S
VDDIO_S
OSCVSS
TESTMODE
EESDA/TMS
EESCL/TCK
IRQ/LATCH1
RUNLED/STATE_RUNLED/E2PSIZE
LINKACTLED1/TDI
LINKACTLED0/TDO
OSCO
OSCI
REG_EN
CLK_25
ERRLED/PME
WAIT_ACK/PME/LATCH0
RST#
NC
NC
NC
NC
SYNC0/LATCH0/PME
SYNC1/LATCH1/PME
GNDANA
GND
LAN9255/ZMX019
81
82 SDA_
83 SCL_
88 PHY_IRQn
89 EEPSIZE
90 LINK1 R3 10K
95 LINK0 R4 10K
1 XTAL_OUT
128 XTAL_IN
13 R5 10K
14 R6 33R CLK_25
17 ERRLED
18
21 RST
RST
48
75
85
86
70 SYNC0
39 SYNC1
15
129
VCC_3V3
SB1
R1 10K
R2 10K
GND
VCC_3V3
GND
GND
GND
GND
VCC_3V3
PHY_IRQn R42 10K
EEPSIZE R7 330R LED1
ERRLED R8 1K LED2
XTAL_IN X1 XTAL_OUT
C27 10pF
C28 10pF
GND
GND
GND
GND
VCC_3V3
R9 10K RST
SW1
C31 0.1uF
GND

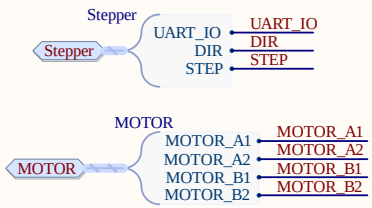

ENC_SSC
ENC_CS
ENC_SCK
ENC_DATA
SW_INT
SWDCLK
SWDIO
SWO
RST
UART
UART_TX
UART_RX
To MCU
CLK_25
PHY_IRQn
Stepper
UART_IO
DIR
STEP

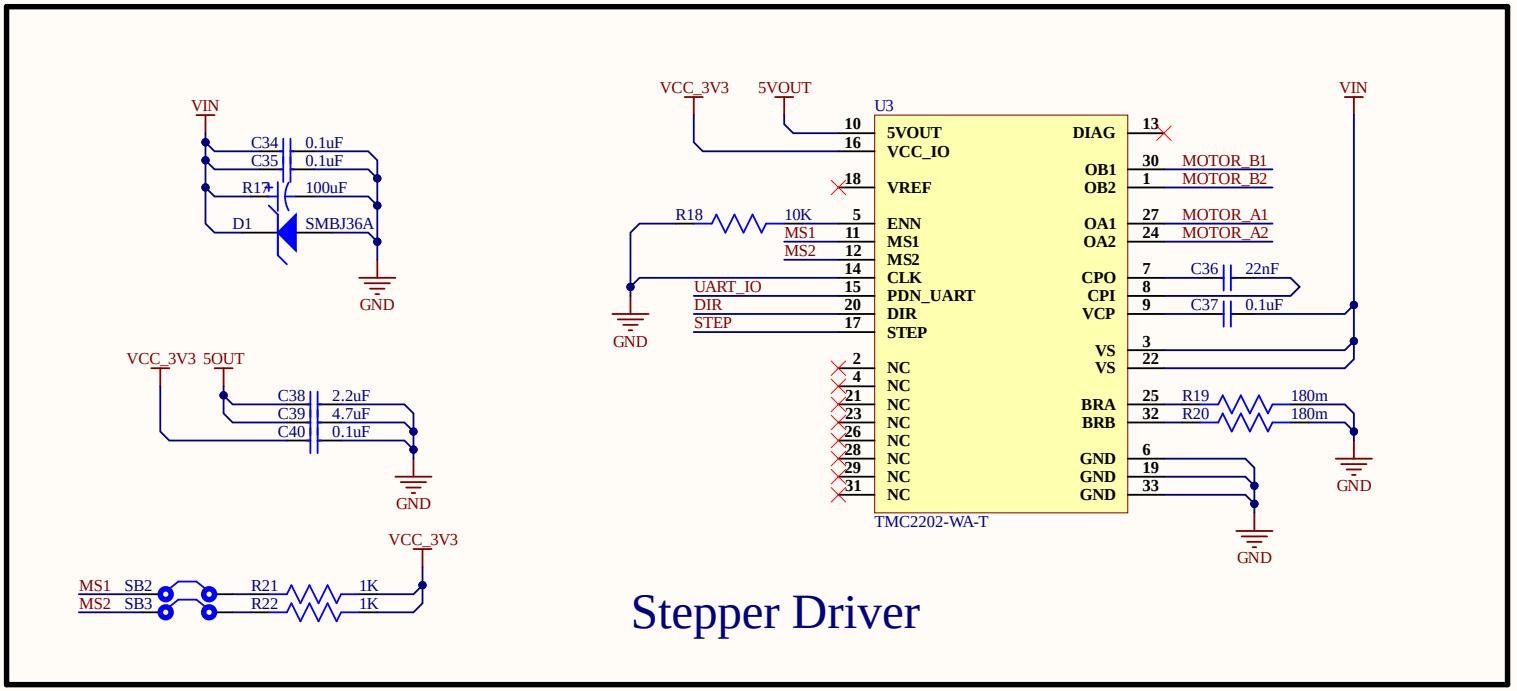

ETH_SIG
TXA_N
TXA_P
RXA_N
RXA_P
TXB_N
TXB_P
RXB_P
RXB_N
I2C
SDA_
SCL_
UART_IO
R10
1K
UART_IO#
U1C
A0/D15/AD15/DIGIO9/GPO9/MII_RXER
A1/ALELO/OE_EXT/MII_CLK25
A2/ALEHI/DIGIO10/GPO10/LINKACTLED2/LEDPOL2
A3/BE0/DIGIO11/GPI11/GPO11/MII_RXDV
A4/BE1/DIGIO12/GPI12/GPO12/MII_RXD0
CS/DIGIO13/GPI13/GPO13/MII_RXD1
WR/ENB/DIGIO14/GPI14/GPO14/MII_RXD2
RD/RD_WR/DIGIO15/GPI15/GPO15/MII_RXD3
TXNA
TXPA
RXNA
RXPA
RXPB
RXNB
TXPB
TXNB
D0/AD0/WD_STATE/SI/SIO0
D1/AD1/EOF/SO/SIO1
D2/AD2/SOF/SIO2
D3/AD3/WD_TRIG/SIO3
D4/AD4/DIGIO3/GPI3/GPO3/MII_LINK
D5/AD5/OUTVALID/SCS#
D6/AD6/DIGIO0/GPI0/GPO0/MII_RXCLK
D7/AD7/DIGIO1/GPI1/GPO1/MII_MDC
D8/AD8/DIGIO2/GPI2/GPO2/MII_MDIO
D9/AD9/LATCH_IN/SCK
D14/AD14/DIGIO8/GPI8/GPO8/MII_TXD3
D13/AD13/DIGIO7/GPI7/GPO7/MII_TXD2
D12/AD12/DIGIO6/GPI6/GPO6/MII_TXD1
D11/AD11/DIGIO5/GPI5/GPO5/MII_TXD0
D10/AD10/DIGIO4/GPI4/GPO4/MII_TXEN
QSPI_SIO0
QSPI_SIO1
QSPI_SIO2
QSPI_SIO3
QSPI_CS
QSPI_CLK
LAN9255/ZMX019
U1B
XIN32
XOUT32
SYNC0
SYNC1
SWDCLK
SWDIO
OPTO_IN
R11
R12
33R
QSPI_CLK
QSPI_CS
ENC_DATA
ENC_SCK
ENC_CS
CLK_25
PHY_IRQn
SWO
VDD_CORE
L6
RST
R13
12.1K
GND

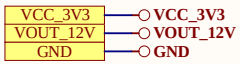

VCC_3V3
GND

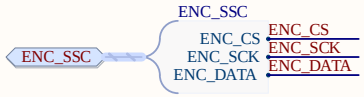

VCC_3V3
R15
R16
10K
SDA_
SCL_
U2
VCC
WP
SDA
SCL
A0
A1
A2
VSS
R14
CAT24C256WI-GT3
GND
EEPROM

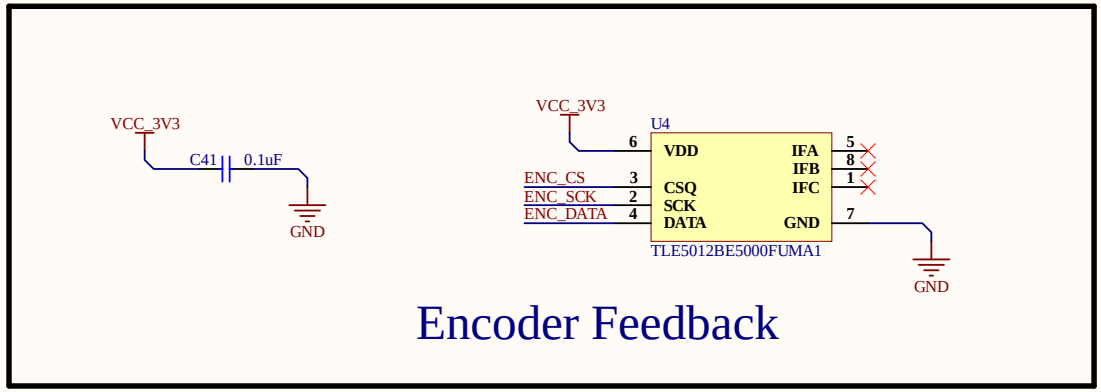

XIN32
X2
XOUT32
C32
C33
10pF
GND
32.768KHz Crystal

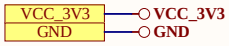

Stepper
Stepper
UART_IO
DIR
STEP
UART_IO
DIR
STEP
MOTOR
MOTOR
MOTOR_A1
MOTOR_A2
MOTOR_B1
MOTOR_B2
MOTOR_A1
MOTOR_A2
MOTOR_B1
MOTOR_B2


VIN
C34 0.1uF
C35 0.1uF
R17 100uF
D1 SMBJ36A
GND
VCC_3V3 5OUT
C38 2.2uF
C39 4.7uF
C40 0.1uF
GND
VCC_3V3
MS1 SB2
MS2 SB3
R21 1K
R22 1K
VCC_3V3 5VOUT
U3
10 5VOUT
16 VCC_IO
18 VREF
R18 10K
5 ENN
11 MS1
12 MS2
14 CLK
UART_IO 15 PDN_UART
DIR 20 DIR
STEP 17 STEP
GND
2 NC
4 NC
21 NC
23 NC
26 NC
28 NC
29 NC
31 NC
DIAG 13
OB1 30 MOTOR_B1
OB2 1 MOTOR_B2
OA1 27 MOTOR_A1
OA2 24 MOTOR_A2
CPO 7 C36 22nF
CPI 8
VCP 9 C37 0.1uF
VS 3
VS 22
BRA 25 R19 180m
BRB 32 R20 180m
GND 6
GND 19
GND 33
TMC2202-WA-T
VIN
GND
GND
Stepper Driver

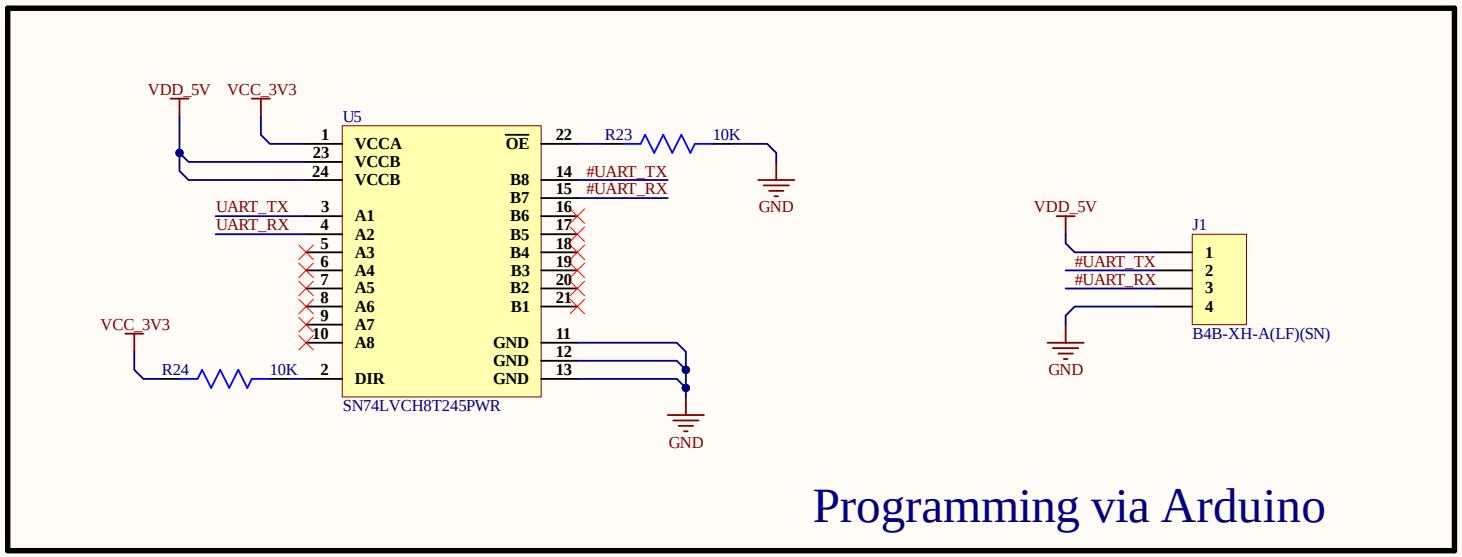

VCC_3V3
VOUT_12V
GND
VCC_3V3
VOUT_12V
GND

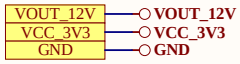

ENC_SSC
ENC_SSC
ENC_CS
ENC_SCK
ENC_DATA
ENC_CS
ENC_SCK
ENC_DATA

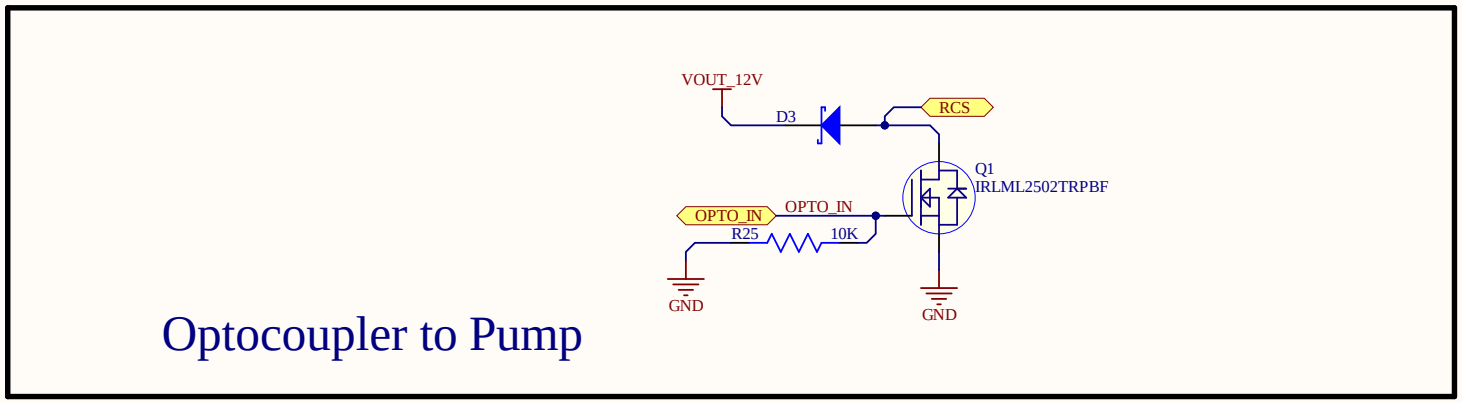

VCC_3V3
C41
0.1uF
GND
VCC_3V3
U4
6
VDD
IFA
5
8
IFB
1
IFC
ENC_CS
3
CSQ
ENC_SCK
2
SCK
ENC_DATA
4
DATA
GND
7
TLE5012BE5000FUMA1
GND
Encoder Feedback

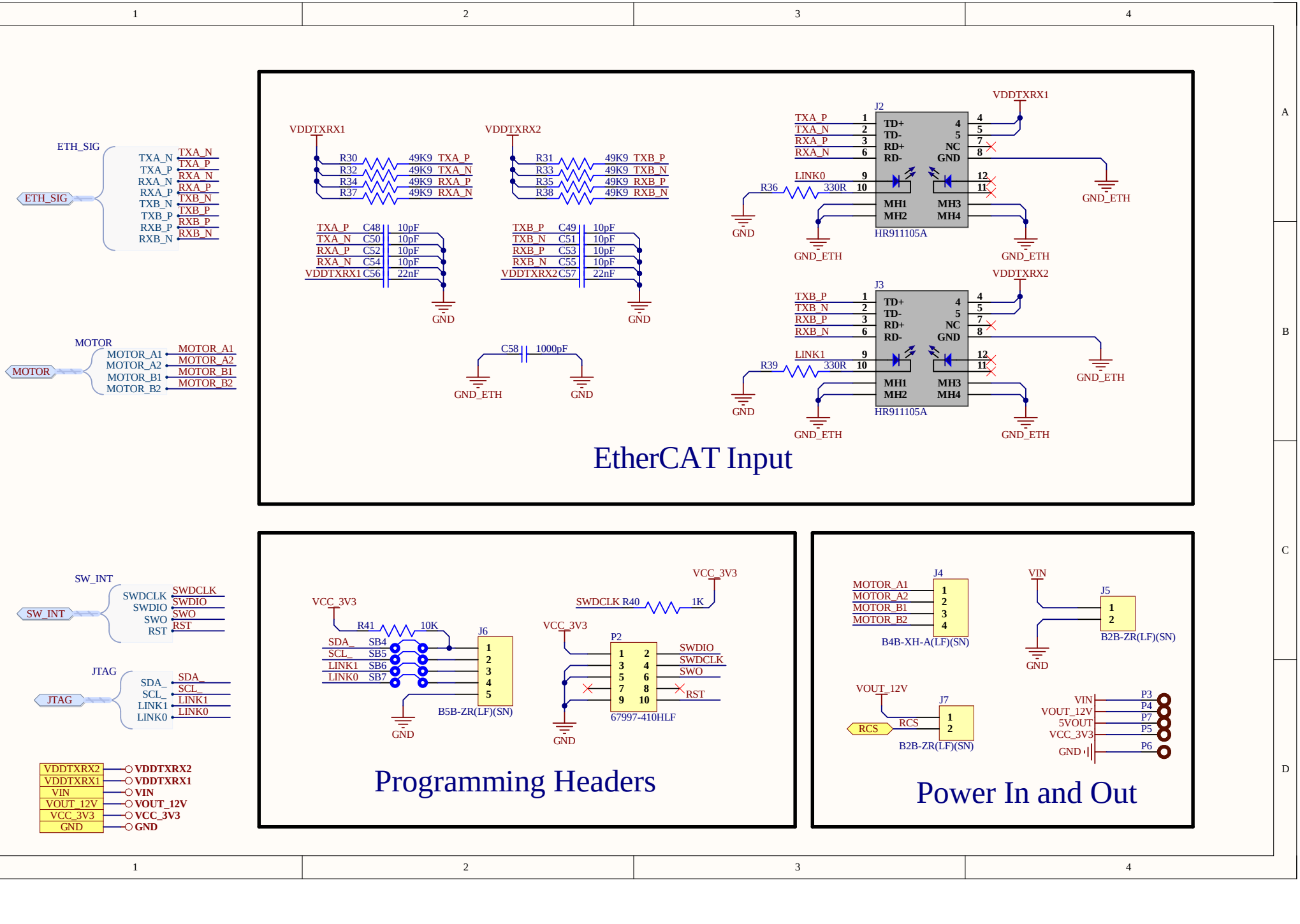

VCC_3V3
GND
VCC_3V3
GND


UART
UART
UART_TX
UART_RX
UART_TX
UART_RX


VDD_5V
VCC_3V3
U5
VCCA
VCCB
VCCB
A1
A2
A3
A4
A5
A6
A7
A8
DIR
OE
B8
B7
B6
B5
B4
B3
B2
B1
GND
GND
GND
SN74LVCH8T245PWR
R23
10K
#UART_TX
#UART_RX
R24
10K
GND
GND
VDD_5V
J1
#UART_TX
#UART_RX
B4B-XH-A(LF)(SN)
GND
Programming via Arduino


VOUT_12V
VCC_3V3
GND
VOUT_12V
VCC_3V3
GND


VOUT_12V
D3
RCS
Q1
IRLML2502TRPBF
OPTO_IN
OPTO_IN
R25
10K
GND
GND
Optocoupler to Pump


ETH_SIG
TXA_N
TXA_P
RXA_N
RXA_P
TXB_N
TXB_P
RXB_P
RXB_N
MOTOR
MOTOR_A1
MOTOR_A2
MOTOR_B1
MOTOR_B2
SW_INT
SWDCLK
SWDIO
SWO
RST
JTAG
SDA_
SCL_
LINK1
LINK0
VDDTXRX2
VDDTXRX1
VIN
VOUT_12V
VCC_3V3
GND
VDDTXRX1
VDDTXRX2
R30 49K9 TXA_P
R32 49K9 TXA_N
R34 49K9 RXA_P
R37 49K9 RXA_N
R31 49K9 TXB_P
R33 49K9 TXB_N
R35 49K9 RXB_P
R38 49K9 RXB_N
TXA_P C48 10pF
TXA_N C50 10pF
RXA_P C52 10pF
RXA_N C54 10pF
VDDTXRX1 C56 22nF
TXB_P C49 10pF
TXB_N C51 10pF
RXB_P C53 10pF
RXB_N C55 10pF
VDDTXRX2 C57 22nF
GND
C58 1000pF
GND_ETH
GND
J2
TXA_P 1
TXA_N 2
RXA_P 3
RXA_N 6
TD+ 4
TD- 5
RD+ NC 7
RD- GND 8
LINK0 9
R36 330R 10
12
11
MH1 MH2 MH3 MH4
HR911105A
GND
GND_ETH
GND_ETH
GND_ETH
VDDTXRX1
J3
TXB_P 1
TXB_N 2
RXB_P 3
RXB_N 6
LINK1 9
R39 330R 10
HR911105A
GND
GND_ETH
GND_ETH
GND_ETH
VDDTXRX2
EtherCAT Input
VCC_3V3
R41 10K
SDA_ SB4
SCL_ SB5
LINK1 SB6
LINK0 SB7
J6
B5B-ZR(LF)(SN)
GND
VCC_3V3
SWDCLK R40 1K
VCC_3V3
P2
SWDIO
SWDCLK
SWO
RST
67997-410HLF
GND
Programming Headers
MOTOR_A1
MOTOR_A2
MOTOR_B1
MOTOR_B2
J4
B4B-XH-A(LF)(SN)
VIN
J5
B2B-ZR(LF)(SN)
GND
VOUT_12V
J7
RCS
RCS
B2B-ZR(LF)(SN)
VIN
VOUT_12V
5VOUT
VCC_3V3
GND
P3
P4
P7
P5
P6
Power In and Out